

1. Consider the following Entities and Relationships

Customer (cust_no, cust_name, address, city)

Loan (loan_no, loan_amt)

Relation between Customer and Loan is Many to many

Constraint: Primary key, loan_amt should be > 0.

Create a Database in 3NF & write queries for following.

1) Find details of all customers whose loan is greater than 2 lakhs.

2) List all customers who have taken at least two loans from bank.

3) List names of all customers in descending order who has taken a loan in pune city.

```
--create table Customer(cust_no integer primary key, cust_name varchar(20),  
                        --address varchar(10), city varchar(10));
```

```
-- insert into Customer values(1, 'Amit', 'Shivajinagar', 'Pune');  
-- insert into Customer values(2, 'Ananya', 'Dadar', 'Mumbai');  
-- insert into Customer values(3, 'Akash', 'Ganjam', 'Berhampur');  
-- insert into Customer values(4, 'Sonali', 'Nehru Nagar', 'Kolkata');  
-- insert into Customer values(5, 'Shubham', 'Ameerpet', 'Hyderabad');  
-- insert into Customer values(6, 'Prakash', 'Hinjewadi', 'Pune');  
-- insert into Customer values(7, 'Rohan', 'Baner', 'Pune');
```

```
--create table Loan(loan_no integer primary key, loan_amt integer);
```

```
--insert into Loan values(101, 1000000);  
--insert into Loan values(102, 500000);  
--insert into Loan values(103, 100000);  
--insert into Loan values(104, 400000);  
--insert into Loan values(105, 200000);  
--insert into Loan values(106, 700000);  
--insert into Loan values(107, 150000);
```

```
--create table CL(cust_no integer references Customer, loan_no references Loan);
```

```
--insert into CL values(1, 105);  
--insert into CL values(2, 104);  
--insert into CL values(3, 102);  
--insert into CL values(4, 101);  
--insert into CL values(5, 103);  
--insert into CL values(6, 106);  
--insert into CL values(7, 107);
```

```
--Find details of all customers whose loan is greater than 2 lakhs.
--select cust_name, address, city, loan_amt from Customer, Loan, CL where
--Customer.cust_no= CL.cust_no and Loan.Loan_no=CL.Loan_no
--and Loan.loan_amt > 200000

--List all customers who have taken at least two loans from bank.
--select cust_name from Customer, Loan, CL where Customer.cust_no= CL.cust_no
--and Loan.loan_no=CL.loan_no group by cust_name having count(Loan.loan_amt)>2

--List names of all customers in descending order who has taken a loan in pune city.
--select cust_name from Customer, Loan, CL where Customer.cust_no= CL.cust_no
--and Loan.loan_no=CL.loan_no and Customer.city = 'Pune' order by cust_name desc
```

2. Consider the following Entities and Relationships
Book (Book_no, title, author, price, year_published)
Customer (cid, cname, addr)

Relation between Book and Customer is Many to Many with quantity as descriptive attribute.

Constraint: Primary key, price should be >0.

Create a Database in 3NF & write queries for following.

- 1) Display author wise details of book.**
- 2) Display all customers who have purchased the books published in the year 2009.**
- 3) Display customer name that has purchased more than 3 books.**

```
--create table Book(Book_no integer primary key, title varchar(30), author varchar(20), price integer,
year_published integer);
```

```
--insert into Book values(101, 'To Kill a Mockingbird', 'Harper Lee', 499, 2002);
--insert into Book values(102, 'Pride and Prejudice', 'Jane Austen', 399, 2006);
--insert into Book values(103, 'The Great Gatsby', 'F. Scott Fitzgerald', 699, 2009);
--insert into Book values(104, 'Harry Potter and the Chamber of Secrets', 'J.K. Rowling', 299, 2009);
--insert into Book values(105, 'Harry Potter and the Deathly Hallows', 'J.K. Rowling', 999, 2012);
--insert into Book values(106, 'The Hobbit', 'J.R.R. Tolkien', 1299, 2015);
--insert into Book values(107, 'Moby Dick', 'Herman Melville', 450, 2009);
```

```
--create table Customer(cid integer primary key, cname varchar(20), addr varchar(20));
```

```
--insert into Customer values(1, 'Anjali', 'Mumbai');
--insert into Customer values(2, 'Ishani', 'Pune');
--insert into Customer values(3, 'Shovna', 'Odisha');
```

```

--insert into Customer values(4, 'Sarthak', 'Bangalore');
--insert into Customer values(5, 'Tushar', 'Kolkata');
--insert into Customer values(6, 'Laksham', 'Delhi');
--insert into Customer values(7, 'Amiya', 'Chennai');

--create table Book_Cust(Book_no integer references Book, cid integer references Customer);

--insert into Book_Cust values(103, 1);
--insert into Book_Cust values(101, 2);
--insert into Book_Cust values(106, 3);
---insert into Book_Cust values(105, 4);
--insert into Book_Cust values(102, 5);
--insert into Book_Cust values(107, 6);
--insert into Book_Cust values(104, 7);

--Display author wise details of book.
--select author, title, Book_no, price, year_published
--from Book order by author;

--Display all customers who have purchased the books published in the year 2009.
--select cname from Book, Customer, Book_Cust where Book.Book_no=Book_Cust.Book_no and
Customer.cid=Book_Cust.cid and Book.year_published = '2009'

--Display customer name that has purchased more than 3 books.
--select cname from Book, Customer, Book_Cust where Book.Book_no=Book_Cust.Book_no and
Customer.cid=Book_Cust.cid group by cname having count(Book.Book_no)>3

```